FANet: An Arbitrary Direction Remote Sensing Object Detection Network Based on Feature Fusion and Angle Classification

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***Abstract*— High-precision remote sensing image object detection has broad application prospects in military defense, disaster emergency, urban planning, and other fields. The arbitrary orientation, dense arrangement, and small size of objects in remote sensing images, however, lead to poor detection accuracy of existing methods. To achieve accurate detection, this article proposes an arbitrary directional remote sensing object detection method, called "FANet," based on feature fusion and angle classification. Initially, the angle prediction branch is introduced, and the circular smooth label (CSL) method is used to transform the angle regression problem into a classification problem, which solves the difficult problem of abrupt changes in the boundaries of the rotating frame while realizing the object frame rotation. Subsequently, to extract robust remote sensing objects, innovatively introduced a pure convolutional model as a backbone network, while Conv is replaced by GSConv to reduce the number of parameters in the model along with ensuring detection accuracy. Finally, the strengthen connection feature pyramid network (SC-FPN) is proposed to redesign the lateral connection part for deep and shallow layer feature fusion and add jump connections between the input and output of the same level feature map to enrich the feature semantic information. In addition, add a variable parameter to the original localization loss function to satisfy the bounding box regression accuracy under different intersection over union (IoU) thresholds, and thus obtain more accurate object detection. The comprehensive experimental results on two public datasets for rotated object detection, a large-scale dataset for object detection in aerial images (DOTA) and HRSC2016, demonstrate the effectiveness of our method.**

***Index Terms*— Angular classification, object detection, oriented bounding box (OBB), remote sensing, strengthen connection feature pyramid network (SC-FPN).**

Manuscript received 17 January 2023; revised 19 March 2023; accepted 1 May 2023. Date of publication 5 May 2023; date of current version 19 May 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61702347 and Grant 62027801, in part by the Natural Science Foundation of Hebei Province under Grant F2022210007 and Grant F2017210161, in part by the Science and Technology Project of Hebei Education Department under Grant ZD2022100 and Grant QN2017132, and in part by the Central Guidance on Local Science and Technology Development Fund under Grant 226Z0501G. *(Corresponding author: Yunzuo Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3273354

I. Introduction

MODERN remote sensing technology has the characteristics of many remote sensing platforms, fast image acquisition, and high image quality. With the shortening of the earth observation period and the significant increase in image resolution, the amount of information contained in remote sensing images has increased rapidly, resulting in the current low utilization of remote sensing images. This has also driven the development of remote sensing image interpretation research. Object detection of remote sensing images is an essential part of remote sensing image processing, which is widely used in urban planning [1], resource surveys [2], and precision agriculture [3]. Therefore, object detection based on remote sensing images has important research significance and application value [4].

Currently, remote sensing image object detection is divided into a two-stage detection method [5] and a one-stage detection method [6]. Two-stage detection methods (e.g., Fast region-convolutional neural network (R-CNN) [8], Faster R-CNN [9], and region-based fully convolutional network (R-FCN) [10]) divide the images into two stages of processing, firstly generating the region candidate frames by region proposal network (RPN) [7], which aims to distinguish the region of interest in the image from the background. And then performing classification and regression operations. These methods have relatively high accuracy but low efficiency due to a large amount of computation required to generate candidate objects and the iterative calculation of features during classification. One-stage algorithms [e.g., SSD [11] and you only look once (YOLO) [12], [13], [14]] generate feature maps through a convolution neural network to obtain the final positioning and classification prediction at once. These methods significantly improve inference speed but compromise precision.

In recent years, the significant improvement of computer processing power and the great success of deep learning technology in applying natural image object detection [9] provides a new impetus and technical approach for remote sensing image object detection. Unlike the objects in the general natural state, remote sensing objects have large-scale differences and many small objects, which brings enormous challenges to the detection task. Remote sensing images are taken from high altitude to the bottom from a top view and

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have high space and coverage characteristics. Among them, the remote sensing objects are arranged in arbitrary directions and have different distribution angles. This is also a challenge in the object detection task of remote sensing images.

To overcome these problems, we propose a new method for object detection in remote sensing images called "FANet." This method can detect nondirectionally aligned objects, while most other localization methods [15], [16] can only detect horizontally or vertically aligned objects. Compared with existing directional bounding box detection methods [17], [18], our method is a pure classification method that does not require additional computation and avoids the typical boundary mutation problem in rotating object detection. Specifically, the main contributions of this article are summarized as follows.

1) We propose a rotating object efficient detection method that uses circular smooth label (CSL) method to determine the angle of the rotating object, thus transforming the angle regression problem into a classification problem to detect objects at any angle more accurately.

2) A method for extracting robust remote sensing object features is proposed, which innovatively uses a pure convolutional model as the backbone network and replaces Conv with GSConv to maintain detection accuracy while reducing the network complexity.

3) A novel module called "strengthened connection feature pyramid network" is proposed to strengthen feature fusion, enrich semantic feature information, and effectively improve the detection abilities of the network for small objects in remote sensing.

4) A variable adaptive loss function is proposed, which meets the bounding box regression accuracy of different levels and improves the detection accuracy of dense objects.

The remained of this article is organized as follows. Section II states the related work of object detection based on deep learning, especially the detection of rotated objects. In Section III, the proposed FANet is presented in detail, and the experiment results are presented to demonstrate the effectiveness of our framework in Section IV. Finally, the conclusion of this article is provided in Section V.

II. Related Work and Problem Analysis

Detection and classification of objects is a fundamental task in computer vision, which has been studied well by many scholars. Although some progress has been made in general-purpose object detection scenarios, it is still challenging in fields such as remote sensing and aerial photography. We first provide an overview of common object detection methods and then introduce object detection methods specifically designed for remote sensing images.

A. General Object Detection Methods

Object detection has a wide range of application scenarios and has recently received extensive attention. Object detection aims to locate specific object instances in an image and mark them with bounding boxes. The label of traditional object detection tasks is horizontal bounding boxes. Most of the early conventional methods use manual features to detect objects. Object detection methods based on standard manual features only involve one or several attributes, such as scale-invariant feature transform (SIFT) [19], histogram of oriented gradients (HOG) [20], and Hough transforms [21]. These methods have minimal generalization ability for multiclass object detection and poor performance. Therefore, a better approach is needed to improve the detection effect of the object.

With the continuous development of deep learning, object detection methods based on convolutional neural networks are endless. The two-stage object detection method is similar to the traditional method but uses a deep learning method to extract deeper high-level semantic features in the feature extraction stage. Among the two-stage object detection methods, the R-CNN method is the most classic, which first selects candidate regions for each image, then extracts features from the object image through a deep convolutional neural network. Finally, it uses the support vector machine (SVM) method to classify these features. The object region pooling layer is added to the Fast R-CNN [8], and the SVM classifier in the R-CNN is changed to a softmax layer, which improves the detection accuracy. Since Fast R-CNN is trained on multi-scale images, although it is more effective than single-scale image object detection methods, the time cost increases significantly. Ren et al. [9] proposed the Faster R-CNN method to solve this problem, which further improved the detection accuracy and speed by using the RPN. Mask R-CNN [22] combines detection and segmentation tasks in one network and employs interpolation operations to align relation oriented inference system (ROIs), resulting in improved detection accuracy.

One-stage object detection methods perform location regression directly on the image. The most classic one-stage algorithms include YOLO series algorithms [12], [13], [14], and SSD [11]. Another iconic algorithm is RetinaNet [23], which proposes focal loss to improve the lack of accuracy caused by the imbalance of positive and negative samples. RetinaNet enables one-stage object detection algorithms to outperform two-stage algorithms in precision and speed. In addition to object detection methods with anchor boxes, many algorithms based on anchor-free can reduce a large number of parameters generated in the process of developing candidate regions. The classic anchor-free algorithm FCOS [24] is based on RetinaNet, which discards the anchor box generation mechanism and adds the concept of centeredness, thereby generating detection boxes directly at the object position on the image. Many key point detection methods are based on anchor-free, including algorithms such as CornerNet [25], CenterNet [26], ExtremeNet [27], and so on. They directly generate detection boxes based on center points or corners, eliminating anchor boxes and improving the detection effect.

These methods have sound detection effects in natural image scenes; however, the conditions of remote sensing image shooting are vastly different from those of natural images. The acquisition mode of remote sensing images is relatively unique. Different shooting angles and distances will increase the difficulty of remote sensing image object detection. Many scholars have done a series of research on the characteristics of objects in remote sensing scenes.

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B. Remote Sensing Image Object Detection Methods

Compared with conventional images, the objects in remote sensing images are directional and arranged in different directions. The traditional horizontal frame detection method has the problem of difficult accurate and compact positioning when dealing with directional objects, especially when detecting objects with large aspect ratios, such as ships and vehicles, and the shape is too narrow. To further improve the positioning ability of objects in the field of remote sensing, rotating object detection has been a broad concern for many scholars.

Rotation region proposal networks (RRPN) [28] proposes an object-oriented detection network based on the framework of R-CNN, which is the first algorithm to introduce a rotation candidate frame to realize object detection in any direction based on the RPN architecture. Rotated region-based CNN (RRCNN) [29] extracts the features of rotating objects by rotating the pooling layer of the region of interest, uses multitask NMS for multicategories, and finally regresses the bounding box of rotating objects. R2PN [30], R3-Net [31], and RRPN algorithms have similar structures. They use specially designed rotation candidate region generation networks to generate candidate regions in any direction. These two methods can effectively detect rotation objects, but the calculation amount is significant. ROI Transformer [32] designed a rotation region of interest learning mechanism, which converts the horizontal region generated by RPN into a rotation region, performs the regression of the horizontal frame and the rotation frame simultaneously, and optimizes the R2PN network for a large amount of calculation. Cao et al. [33] proposed a multiangle rotating object detection algorithm for remote sensing images by improving the feature pyramid network, which improved the performance of remote sensing small object detection.

Remote sensing images are too densely distributed in some special situations, such as ships in ports, aircraft in airports, vehicles in parking lots, etc. Densely distributed objects are prone to feature aggregation, leading to false and missed detection. Long et al. [34] combine deep learning methods with various traditional ways, effectively detecting high-density objects in complex backgrounds, but the process is very complicated. In order to improve the discrimination of different objects, many methods use feature enhancement. SCRDet [18] achieves feature enhancement of dense objects by increasing channel and pixel attention. SCRDet++ [35] creatively applies the idea of denoising to object detection and uses the intermediate features of the semantic segmentation network to guide the feature extraction process of detection, thereby improving the boundary discrimination degree of dense objects. R3Det [36] designed a feature thinning module to realize feature reconstruction and alignment and improve the positioning accuracy of dense objects. These feature enhancement methods all introduce additional operation modules, which will, however, enhance the accuracy of dense object detection.

The common parameter representation will accompany the typical boundary mutation problem when the rotating frame coordinate regression is carried out. The boundary mutation problem is the unique characteristic of a rotating frame. To solve this problem, adaptive period embedding (APE) [37] uses a pair of periodic trigonometric functions to characterize angle information, avoiding boundary mutation. For the improvement of the loss function, SCRDet [18] designed Smooth L1 Loss, which is specially used to solve the boundary problem of rotating boundary box regression. SCRDet++ [35] adds a new intersection over union (IoU) constant factor based on Smooth L1 Loss to further unravel the boundary problem of the rotating frame. AFC-Net [38] proposed the oriented bounding box (OBB) multidefinition and selection strategy to regress the rotation boundary box more smoothly. Modulated loss [39] calculates the loss function on the boundary using all sudden change values and selects the minimum loss value for training to mitigate the impact of the boundary problem. The periodicity of the angle in these methods, however, quickly affects the detection accuracy and does not entirely solve the boundary problem from the root. Yang and Yan [40] designed one-hot coding and CSL considering the commutativity of edges and the periodicity of angles, increased the tolerance error between adjacent angles, and converted the regression problem of angles into a classification problem to adapt to the periodicity of angles. The algorithm of this article draws on the idea of CSL and proposes the FANet method on this basis.

III. Method

This article proposes an arbitrary direction remote sensing object method (FANet) based on feature fusion and angle classification. The overall framework of this method is shown in Fig. 1.

A. Overall Framework

The pure convolution model ConvNext-Tiny [41] is used in the backbone network of this article. Where $C = (96, 192, 384, 768)$, $B = (3, 3, 9, 3)$, C represents the number of input channels, and B represents the number of times to stack ConvNext block modules repeatedly. ConvNeXt uses transformer's [42] design components to improve the classic CNN. GSConv is introduced into the neck layer of the network model. On this basis, a cross-level partial network module VOV-GS cross stage partial network (VOV-GSCSP) is designed using the one-time aggregation method to reduce the network model parameters. The combination structure of strengthen connection feature pyramid network (SC-FPN) and pyramid attention network (PAN) [43] is introduced to achieve better feature fusion. SC-FPN obtains feature maps for prediction through a bottom-up structure and a top-down structure, while PAN conveys robust positioning features from the bottom-up. The composite structure aggregates the parameters of different backbone layers to different detection layers that are used to detect remote sensing image objects so that the network can detect large, medium, and small-scale objects.

In this article, FANet uses regression loss, object confidence loss, classification loss and angle classification loss, respectively. The object confidence loss, classification loss, and angle classification loss are all calculated by binary cross entropy loss (BCE With Logits Loss), while the regression loss is based on complete IoU (CIoU) by introducing power transformation, and alpha-CIoU loss is used to improve the accuracy of bounding box regression. In postprocessing, the rotation

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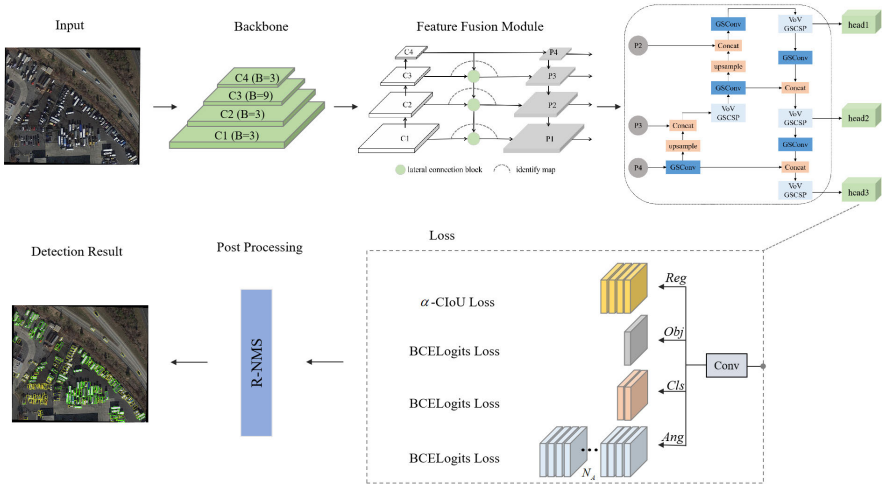


Fig. 1. Architecture of FANet.

nonmaximum suppression method eliminates the redundant prediction frame, and the final detection result is obtained.

B. Backbone Network

FANet uses ConvNeXt block to extract robust remote sensing object features and classify objects. In ConvNeXt, the architecture of standard CNN is modernized into the structure of a hierarchical visual converter. The ConvNeXt block contains 7 × 7 depthwise convolution, 1 × 1 convolution, and the nonlinear GELU activation function. Whereas in 1 × 1 convolution layer uses LayerNorm. For input z, GELU can be expressed as

$$\mathrm{GELU}(\mathrm{z}) = \mathrm{z}P(\mathrm{Z} \le \mathrm{z}) = \mathrm{z}\,\Phi(\mathrm{z}) = \mathrm{z} \cdot \frac{1}{2}\left[1 + \mathrm{erf}\left(z/\sqrt{2}\right)\right]. \tag{1}$$

It is important to note that the GELU expression mentioned in (1) can be approximated as

$$0.5x\left(1 + \tanh\left(\sqrt{2/\pi}\left(x + 0.044715x^3\right)\right)\right). \tag{2}$$

In the ConvNeXt model, the LayerNorm method is employed to avoid the drawbacks of the batch normalization technique widely used in existing deep CNN architectures. For all hidden cells in the same layer, the LayerNorm statistics are calculated as

$$\mu^l = \frac{1}{H}\sum_{i=1}^{H} a_i^l \sigma^l = \sqrt{\frac{1}{H}\sum_{i=1}^{H}\left(a_i^l - \mu^l\right)^2} \tag{3}$$

where H represents the number of hidden cells in each layer.

In addition, the ConvNeXt adds a separate down-sampling layer between phases, which uses a 2 × 2 convolution layer with a step length of 2.

C. Structural Design of Neck Layer

The backbone network ConvNeXt can extract robust remote sensing object features. Still, it will also increase the number of model parameters, thus reducing the speed of remote sensing image object detection; however, the lightweight model constructed by many deep separable convolution layers cannot achieve sufficient accuracy. Therefore, GSConv [44] is introduced to reduce the complexity of the model and maintain its accuracy. The DWConv [45] is used in the GSConv module.

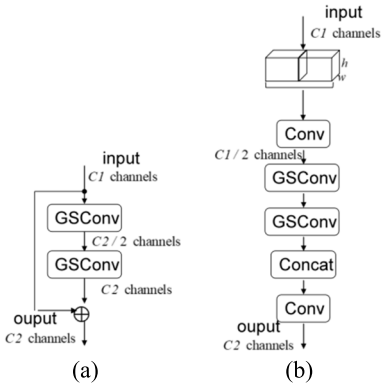


Fig. 2. Structural diagrams of two models structure diagram of (a) GS bottleneck model and (b) VOV-GSCSP model.

For this purpose, FANet introduced GSConv in the neck layer and designed the cross-level partial network module VoV-GSCSP based on it using a one-time aggregation method. First, the lightweight convolution method GSConv is used to replace Conv. Then, based on GSConv, the GS bottleneck is introduced. Fig. 2(a) shows the structure of the GS bottleneck module. The VOV-GSCSP module reduces the complexity of calculation and network structure but maintains sufficient accuracy. Its design is shown in Fig. 2(b).

If GSConv is used in all phases of the model, the network layers of the model will be deeper, and the deep layers will exacerbate the resistance to data flow and significantly increase the inference time. When these feature maps reach the neck layer, they become slender (the channel dimension reaches the maximum, and the width and height dimension reaches the minimum), and no transformation is required. Therefore, a better choice is to use GSConv only in the neck layer. Fig. 3 shows the neck layer structure in the FANet framework.

D. Strengthen Connection Feature Pyramid Network

The feature pyramid network includes three parts: 1) as shown on the left side of Fig. 4, the line from shallow layer to deep layer; 2) as shown on the right side of Fig. 4, the line

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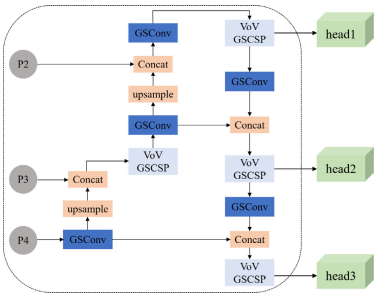


Fig. 3. Network structure of neck layer in FANet framework.

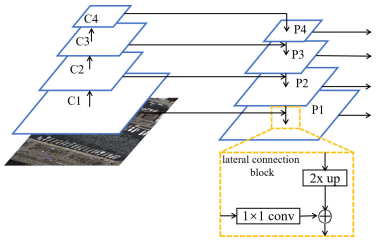


Fig. 4. FPN structure diagram.

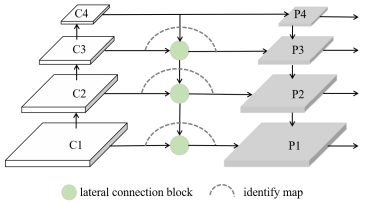


Fig. 5. SC-FPN structure diagram.

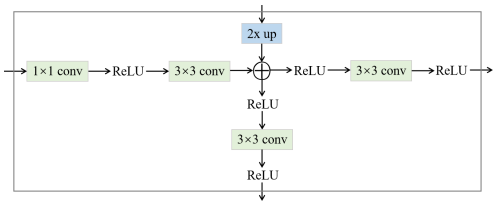


Fig. 6. Schematic of the lateral connection structure of SC-FPN.

from deep layer to shallow layer; and 3) lateral connection, as shown in the yellow dashed box in Fig. 4.

The lateral connection structure of feature pyramid networks (FPN) is too simple, and the deep and shallow features are, however, not fully utilized. To make it more suitable for small object detection in remote sensing images, this article proposes a strengthened connection feature pyramid network. The overall architecture is shown in Fig. 5. SC-FPN improves the lateral connection part to better fuse the deep and shallow features and adds a jump connection between the lateral input and output of the same layer to fuse more features without additional consumption.

The lateral connection structure of SC-FPN is shown in Fig. 6. For the lateral input shallow characteristic convolution, rectified linear unit (ReLU) activation function, and 3 × 3 convolution extract shallow texture features. The purpose of 1 × 1 convolution is to reduce the channel numbers of the lateral input characteristic map and the amount of calculation. The purpose of 3 × 3 convolution is to extract texture features from the lateral input shallow feature map again. Then, as soon as FPN, the deep and shallow feature maps are fused through element addition operation to strengthen the features further. After feature fusion, two ReLU activation functions and a 3 × 3 convolution perform secondary feature extraction on the fused feature map to ensure the stability of the fused features. In addition, inspired by residual learning, SC-FPN also adds an identity map between the lateral input and output of each lateral connection to fuse more features without adding any additional calculation, which is conducive to subsequent object detection and recognition.

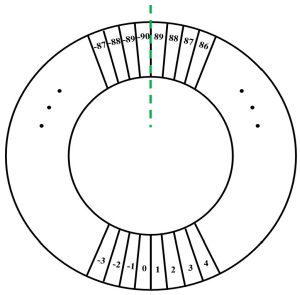


Fig. 7. CSL schematic.

E. Angle Prediction Branch

The objects in remote sensing images are arranged in different directions and are densely distributed. By adding angle prediction branches, the objects at any angle can be detected more accurately. Most regression-based rotation detection methods use five tuples to represent the directional bounding box of the remote sensing object, where (c_x, c_y) represents the central point coordinates of the bounding box, w and h, respectively, represent the width and height of the object, and θ represents the rotation angle of the object.

Since objects with relatively large lengths and widths are susceptible to angle changes, such methods, however, suffer from inconsistent angle regression in the boundary case. It makes the loss value of the model at the boundary suddenly increase, which leads to the low detection accuracy of such detection methods in the boundary conditions [18].

Based on this problem, this article uses the rotation frame detection method based on the CSL, as shown in Fig. 7. By transforming the regression problem of angle into a classification problem to solve the recurring problem of angle and then solve the boundary discontinuity problem to realize the accurate localization of remote sensing image objects.

The CSL converts the angle prediction of the regression task into a high-precision classification task. A periodic cyclic label coding based on high-precision angle classification is designed for rotation detection. By setting a window function, the model can measure the angular distance between the predicted and actual labels on the ground. The equation of CSL is

$$\mathrm{CSL}(x) = \begin{cases} g(x), & \theta - r < x < \theta + r \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Among them, $g(x)$ is the window function. This article uses the Gaussian function to satisfy symmetry and periodicity.

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F. Loss Function

The loss function calculation of the FANet is shown in (5), where L_{Obj}, L_{Cls}, L_{Reg} and L_{Ang}, respectively, represent confidence loss, classification loss, regression loss, and angle classification loss. Hyperparameters λ_1, λ_2, λ_3 and λ_4 are used to control balance. The default setting is {1, 1, 1, 1}

$$L = \lambda_1 L_{\mathrm{Obj}} + \lambda_2 L_{\mathrm{Cls}} + \lambda_3 L_{\mathrm{Reg}} + \lambda_4 L_{\mathrm{Ang}} \tag{5}$$

where L_{Obj}, L_{Cls} and L_{Ang} re calculated by BCE.

We use $\alpha - \mathrm{CIoU}$ loss to correct the positioning loss, improve the accuracy of bounding box regression, and obtain higher quality anchors. $\alpha - \mathrm{CIoU}$ loss is defined as follows:

$$L_{\alpha-\mathrm{CIoU}} = 1 - IoU^{\alpha} + \left| \frac{\rho^{2\alpha}\left(b, b^{gt}\right)}{c^{2\alpha}} \right| + (\beta v)^{\alpha}. \tag{6}$$

The power parameter alpha can be used as a hyperparameter to adjust the alpha-IoU loss to meet different levels of bounding box regression accuracy. It can be seen that when alpha >1, the loss will decrease with the increase of IoU, which improves the model convergence speed and is more conducive to the object with a larger IoU.

The results were analyzed by ablation experiments. In this article, α is set to 3. The specific implementation will be described in detail in the ablation experiment section.

IV. Experiment and Results

We conduct extensive experiments on two popular and challenging benchmarks, a large-scale dataset for object detection in aerial images (DOTA) [46] and HRSC2016 [47], to verify the effectiveness and superiority of our method. The details of these two datasets are presented in Section III-A, the exact details of the experimental setup are given in Section III-B, and the ablation experiments and detailed analysis sections are presented in Section III-C. The comparison results of different methods on the two datasets are elaborated in Section III-D.

A. Datasets

1) DOTA Dataset: DOTA [46] is a large-scale dataset for object detection in remote sensing images. It contains 2806 aerial images of different sizes from 800 × 800 to 4000 × 4000 pixels. These images are divided into 15 categories, namely plane (PL), ship (SH), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GTF), harbor (HA), bridge (BR), small vehicle (SV), large vehicle (LV), helicopter (HC), roundabout (RA), soccer ball field (SBF), and swimming pool (SP). Each sample is marked with an arbitrary quadrilateral, in which small-sized instances (10–50 pixels) and medium-sized instances (50–300 pixels) are in the majority.

2) HRSC2016 Dataset: HRSC2016 [47] is a high-resolution image dataset for ship detection. The objects in this dataset have the characteristics of large aspect ratio and arbitrary direction arrangement, and the image size is between 300 × 300 and 1500 × 900. There are 1061 images in the dataset, including 436, 181, and 444 for training, verification, and testing.

B. Implementation Details

Due to the large resolution of remote sensing images, it is easy to cause a sudden increase in video memory during the training or prediction process. At present, there are generally two solutions. One is to reduce the image's resolution, such as resizing an image with a size of 5000 × 5000 to 1000 × 1000, but this will lose image information. Especially the image datasets such as DOTA, which contain many small objects. If this method is adopted, the size of the small objects will be further reduced, and their features may disappear during the layer-by-layer extraction process of the deep network, which is not conducive to the detection of small objects. The second is to segment the image. We used this method during the experiment. Specifically, the image is divided into 1024 × 1024 images, and the cutting overlap rate is set to 20%. The images smaller than 1024 × 1024 are filled with black. In this way, the loss of object information at the cutting edge can be avoided while ensuring image resolution. Whereas in the HRSC2016 dataset, the ships are of moderate size, and we scale the images to 640 × 640 during training.

In this article, all experiments are implemented on PyTorch, using Intel(R) Xeon(R) Gold 6226R CPU @ 2.90 GHz, NVIDIA Tesla V100S-PCIE-32 GB GPU, and the operating system is CentOS7. The network uses the stochastic gradient descent (SGD) algorithm as the optimizer, and the initial learning rate, momentum, and weight decay are set to 0.01, 0.937, and 0.0005, respectively. In this article, we use a series of data augmentation methods in the DOTA experimental dataset, including random rotation, translation, and scaling of images. The data augmentation method is used to improve the robustness and generalization ability of the network model.

C. Ablation Study

To demonstrate the effectiveness of the proposed method, we performed several ablation experiments on the DOTA dataset. The effects of different combinations of five factors, including backbone network, neck layer design, SC-FPN, boundary regression loss (BRL), and CSL, on the final experimental results are considered. We use FPS, recall, precision, F1-score, mAP, parameter, and giga floating-point operations per second (GFLOPs) as evaluation criteria to verify the effectiveness of our method, and the results are shown in Table I.

1) Backbone Network: We compared the results of CSP-Darknet and ConvNeXt as the backbone network for remote sensing image object detection. As shown in Table I, after replacing the backbone network from CSPDarknet to ConvNeXt, the mAP on the DOTA dataset increased from 70.93% to 72.21%, an improvement of 1.28%. ConvNeXt is to introduce the latest part of transformer network ideas and techniques into the existing modules of the CNN network, thus combining the advantages of both networks and improving the performance of the CNN network. ConvNeXt can extract robust remote sensing object features, but it also increases the number of model parameters, thus reducing the speed of remote sensing image object detection.

2) Circular Smooth Label: As can be seen from the results of the second and third sets of experiments in Table I, the

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TABLE I
ABLATION EXPERIMENTS FOR EACH COMPONENT ON DOTA DATASET

Backbone	BRL	CSL	SC-FPN	Neck-Conv	Neck-GSConv	FPS/(frame·s^{-1})	recall (%)	precision (%)	F1-score	mAP(%)	Parameter	GFLOPs
CSPDarknet	CIoU					29.52	69.46	68.38	0.6892	70.93	24M	53.9
ConvNeXt	CIoU					17.43	71.84	70.12	0.7097	72.21	87M	90.1
ConvNeXt	CIoU	√				15.84	72.13	73.06	0.7259	73.65	89M	91.4
ConvNeXt	CIoU	√	√			15.16	75.68	72.91	0.7427	75.83	91M	93.8
ConvNeXt	α-CIoU	√	√			16.27	76.44	73.82	0.7511	76.74	94M	94.1
ConvNeXt	α-CIoU	√	√	√		17.11	76.53	73.65	0.7506	76.59	96M	95.3
ConvNeXt	α-CIoU	√	√		√	26.38	76.51	73.64	0.7505	76.57	64M	72.3

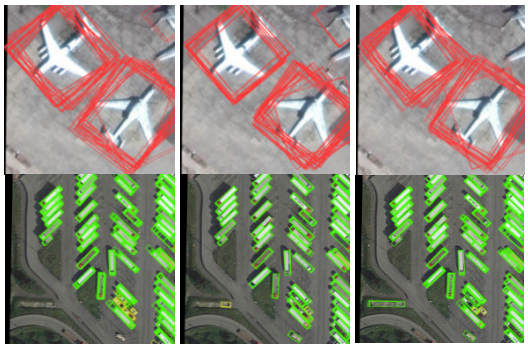
TABLE II
COMPARISON OF DETECTION PERFORMANCE AT DIFFERENT WINDOW RADIUS

mAP/%			
$r=0$	$r=2$	$r=4$	$r=6$
70.56	73.65	72.11	71.47

mAP improved from 72.21% to 73.65% after adding the CSL angle prediction branch to the second set of experiments, an improvement of 1.44%. With CSL, the angle prediction can be transformed from regression to a classification problem, thus solving the angle periodicity problem. The window radius in the CSL method is one of the critical parameters that affect the overall model performance. When the radius of the window function is too large, the discrimination of the prediction results is too small. When the window radius is too small, the window function tends to be an impulse function and cannot learn the direction and scale information. For this problem, suitable radii from 0 to 6 are chosen for the experiments, and the detection performance of the model at different radius sizes is shown in Table II. It can be seen that the best detection performance on the DOTA dataset is achieved when $r = 2$.

*3) SC-FPN:*Using the SC-FPN based on the third group of experiments, the mAP is improved from 73.65% to 75.83%, an improvement of 2.18%. To solve the problem of a simple horizontal connection structure of FPN and insufficient utilization of deep and shallow features, the SC-FPN is proposed to extract shallow features for fusion and perform secondary feature extraction to ensure the stability of the fused features. In addition, jump connections are added between the lateral inputs and outputs of the same layer to fuse more features without additional consumption, further enriching the semantic information of the features.

4) Boundary Regression Loss: Based on the fourth set of ablation experiments, we analyze the role of the BRL $\alpha-$CIoU by setting the value of αThe parameter α can satisfy the accuracy of the bounding box regression under different IoU thresholds to obtain more accurate object detection.



(a) (b) (c)

Fig. 8. Bounding box regression effects for different α values in the DOTA dataset. (a) $\alpha = 2$, (b) $\alpha = 3$, and (c) $\alpha = 4$.

TABLE III
EFFECT OF DIFFERENT α VALUES ON DETECTION PERFORMANCE

Value of α	mAP
α=1.5	75.45%
α=2	76.26%
α=2.5	76.30%
α=3	76.57%
α=3.5	75.63%
α=4	75.44%

It can be seen that the best detection is achieved on both datasets when $\alpha = 3$. The different ways of locating the bounding boxes in the DOTA dataset when $\alpha = 2$, $\alpha = 3$ and $\alpha = 4$ are shown in Fig. 8, and it is easy to see that the most accurate bounding box regression is obtained when $\alpha = 3$. Therefore, in this experiment, $\alpha = 3$ is selected.

1) *Neck Layer Design:*After changing Conv in Neck to GSConv, the number of model parameters decreased by 33.3%, the amount of calculation decreased by 24.1%, and the amount of mAP decreased by 0.02%. The GSConv module maintained the detection accuracy

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TABLE IV
PERFORMANCE COMPARISON RESULTS OF OBB TASK ON DOTA DATASET. THE BEST RESULT IN EACH CATEGORY IS HIGHLIGHTED IN BOLD

Methods	mAP	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
RRPN [28]	61.01	88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58
RADet [48]	69.09	79.45	76.99	48.05	65.83	65.46	74.40	68.86	89.70	78.14	74.97	49.92	64.63	66.14	71.58	62.16
RoI Trans. [32]	69.56	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67
DRN [49]	70.70	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50
O2-DNet [50]	71.04	89.31	82.14	47.33	61.21	71.32	74.03	78.62	90.76	82.23	81.36	60.93	60.17	58.21	66.98	61.03
R^3Det [36]	71.69	89.54	81.99	48.46	62.52	70.48	74.29	77.54	90.80	81.39	83.54	61.97	59.82	65.44	67.46	60.05
SCRDet [18]	72.61	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	**86.86**	65.02	**66.68**	66.25	68.24	65.21
CFC-Net [51]	73.50	89.08	80.41	52.41	70.02	76.28	78.11	87.21	90.89	84.47	85.64	60.51	61.52	67.82	68.02	50.09
APE [37]	75.75	89.96	83.62	53.42	**76.03**	74.01	77.16	79.45	90.83	87.15	84.51	**67.72**	60.33	74.61	71.84	65.55
DODet [52]	75.89	89.61	83.10	51.43	72.02	79.16	81.99	87.71	**90.89**	86.53	84.56	62.21	65.38	71.98	70.79	61.93
MRDet [53]	76.24	89.49	84.29	55.40	66.68	76.27	82.13	87.86	90.81	86.92	85.00	52.34	65.98	**76.22**	**76.78**	**67.49**
Ours	**76.57**	**91.24**	**84.42**	**56.53**	68.12	**79.82**	**83.48**	**88.91**	90.56	85.72	86.35	61.96	64.85	74.08	69.14	63.35

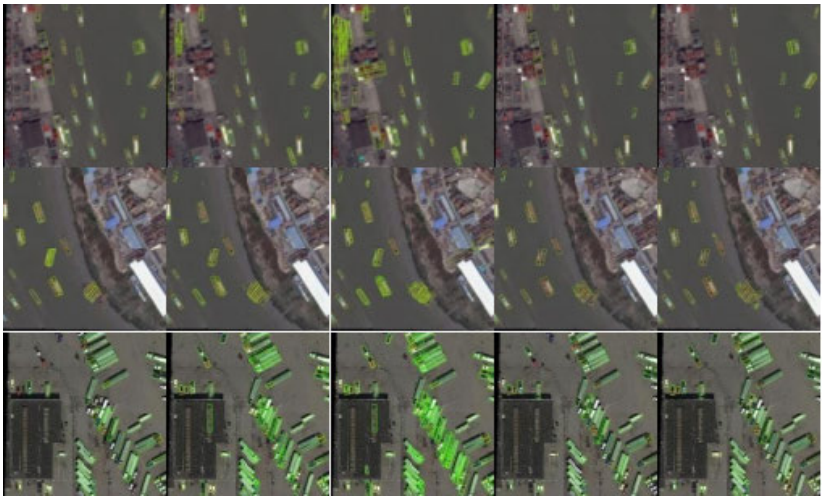


Fig. 9. Comparison of visualization results of different algorithms on DOTA dataset. (a) DRN, (b) O2-DNet, (c) CFC-Net, (d) MRDet, and (e) FANet.

while reducing the computational and network structure complexity.

D. Comparisons With State of the Arts

In this section, we compare our method with the state-of-the-art methods on DOTA and HRSC2016.

1) *Results on DOTA:* Table IV presents the performance of our method and the state-of-the-art methods on the OBB task of the DOTA test set. Our method reaches the peak with an mAP of 76.57%, which is higher than the previous best result (MRDet [53]) by about 0.4%. Moreover, FANet achieves progress on some categories, such as vehicle and ship, whose objects are often distributed in a high density with similar scales and orientations. For SVs, LVs and ships, FANet achieves 79.82%, 83.48%, and 88.91% in mAP with 0.66%, 1.35%, and 1.05% improvements than the second-best models (79.16%, 82.13%, and 87.86%), respectively. In addition, the proposed method achieves accurate detection of objects with different alignment directions. For airplanes, FANet achieves 1.26% mAP improvements than the SCRDet algorithm (89.98%).

In Fig. 9, we show the visualization results of our method and other comparison methods. We can see that our method performs well on aerial objects in any direction, even in dense scenes. We attribute this to using a backbone network with better feature extraction capability and SC-FPN with better feature fusion capability. More importantly, we convert the sensitive angle regression task into a pure classification

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TABLE V
PERFORMANCE COMPARISON RESULTS OF OBB TASK ON HRSC2016 DATASET. THE BEST RESULT IN EACH CATEGORY IS HIGHLIGHTED IN BOLD

Methods	RRPN [28]	R^2PN [30]	RRD [54]	RoI Trans.[32]	SKNet [55]	R^3Det [36]	CFC-Net [51]	MRDet [53]	DODet [52]	**Ours**
Backbone	ResNeXt-101	VGG16	VGG16	ResNet-101	Hourglass-104	ResNet-101	ResNet-101	ResNet-101	ResNet-101	ConvNeXt
AP(%)	79.05	79.60	84.30	86.20	88.30	89.26	89.70	89.94	90.89	**91.10**



Fig. 10. Detection results of HRSC2016 dataset.

problem, which solves the angle periodicity problem and makes the object localization more accurate. In addition, the parameter added to the localization loss function improves the accuracy of the enclosing frame regression. The joint action completes the accurate remote sensing object detection. Our method casts the overall task of object detection to a classification problem that does not add ambiguity or complexity to deal with oriented boxes. The results show that our method ranks first among the other methods in OBB detection in six categories.

2) *Results on HRSC2016:* HRSC2016 contains many remote sensing-oriented ships with large aspect ratios, large scale, and extensive orientation. Our method achieves good performance on the HRSC2016 dataset. FANet can automatically learn anchor dimensions based on the labels of the dataset. Table V depicts the comparison performance between our FANet with several state-of-art methods on the HRSC2016 test set. It can be seen that our method outperforms all other methods evaluated on this dataset. It specifically achieves 91.10 AP, which is 1.26% higher than the second-best method, which is R^3Det [36], with 89.26 AP. Some results on HRSC2016 are shown in Fig. 10, which can be seen that our method detects high-quality objects in different orientations.

V. CONCLUSION

The detection of directional objects in remote sensing images is a challenging task. In this article, we have presented an effective detection method for oriented and densely packed objects in aerial images called "FANet." We propose a method for extracting robust remote sensing object features. To improve the detection performance of densely distributed objects, we designed a novel feature fusion module. In addition, the angular classification module, instead of the angular regression module is adapted to the detection framework for the first time. This module can obtain the direction information of objects more accurately without the problem of boundary discontinuity. To meet the bounding box regression accuracy of different levels, we innovatively add a variable parameter in the positioning loss function.

Our method not only achieves state-of-the-art performance on popular datasets in aerial images (i.e., DOTA and HRSC2016) but also outperforms some classes with dense-packed and orientation variation objects. We verified the proposed FANet in two datasets. In the experiments, we compared and analyzed the performance of the proposed algorithm with popular object detection methods. In HRSC2016, the proposed FANet achieved 91.24% in AP. Meanwhile, in the more challenging dataset, DOTA, FANet has also achieved SOTA performance, which reaches 76.57% in mAP. The ablation experiment further verified the effectiveness of the above modules proposed in FANet. Empirical studies show that FANet achieves state-of-the-art detection performance. In the future, we will explore a more efficient and accurate method for detecting oriented objects in aerial images.

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